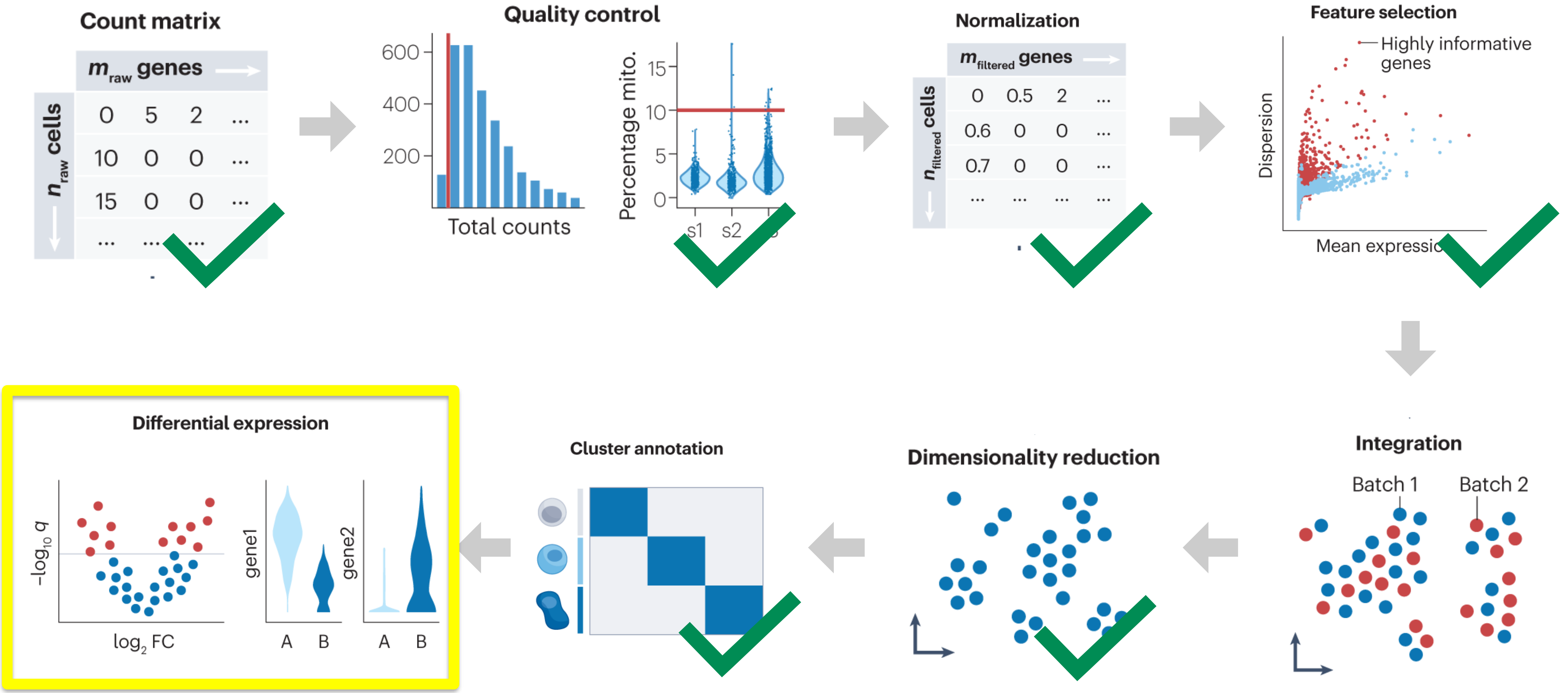
Single Dell Data Analysis Course

Differential abundance and gene expression analysis

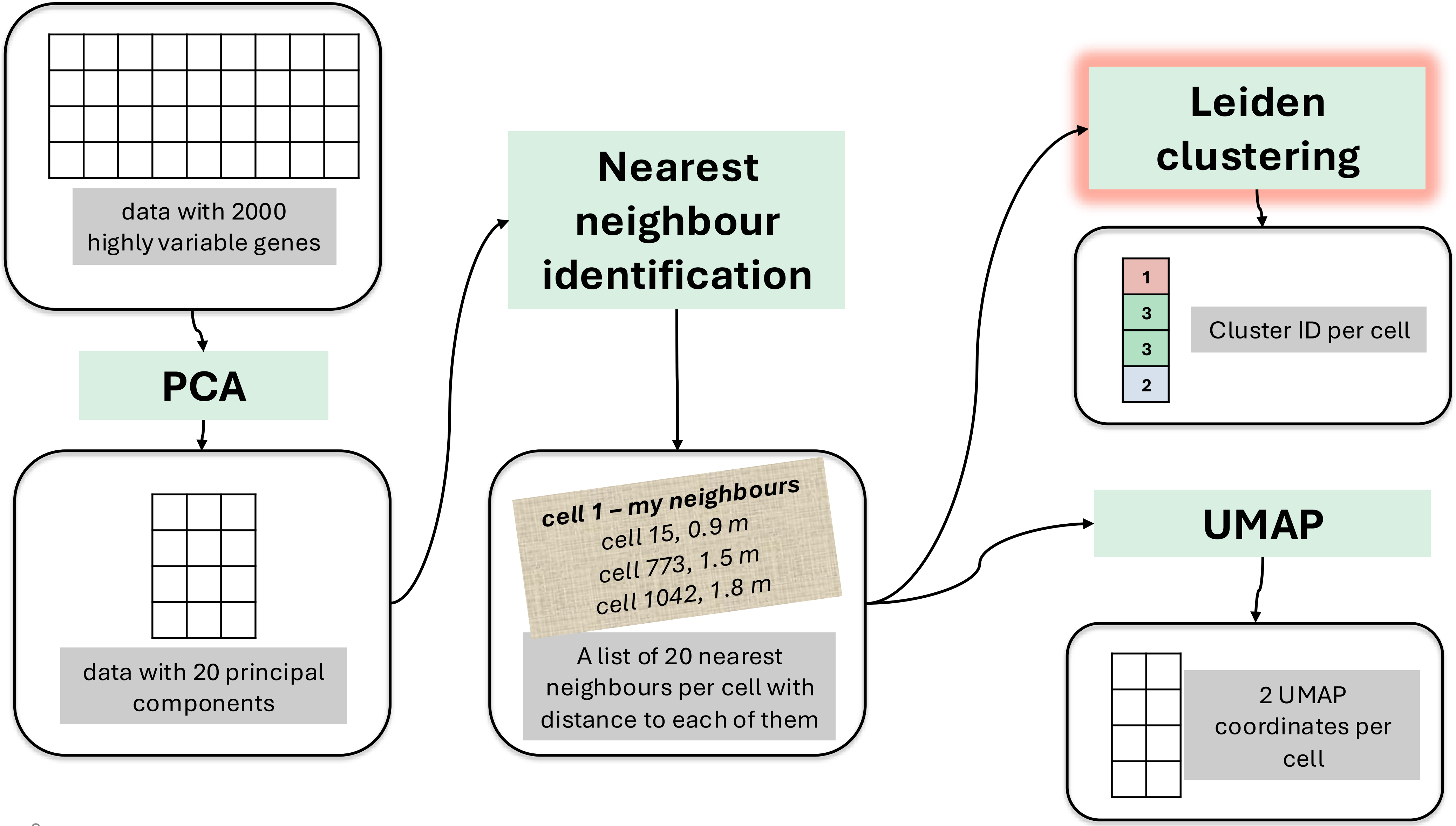
Lisa Buchauer
Professor of Systems Biology of Infectious Diseases
Department of Infectious Diseases and Intensive Care
Charité - Universitätsmedizin Berlin

Today

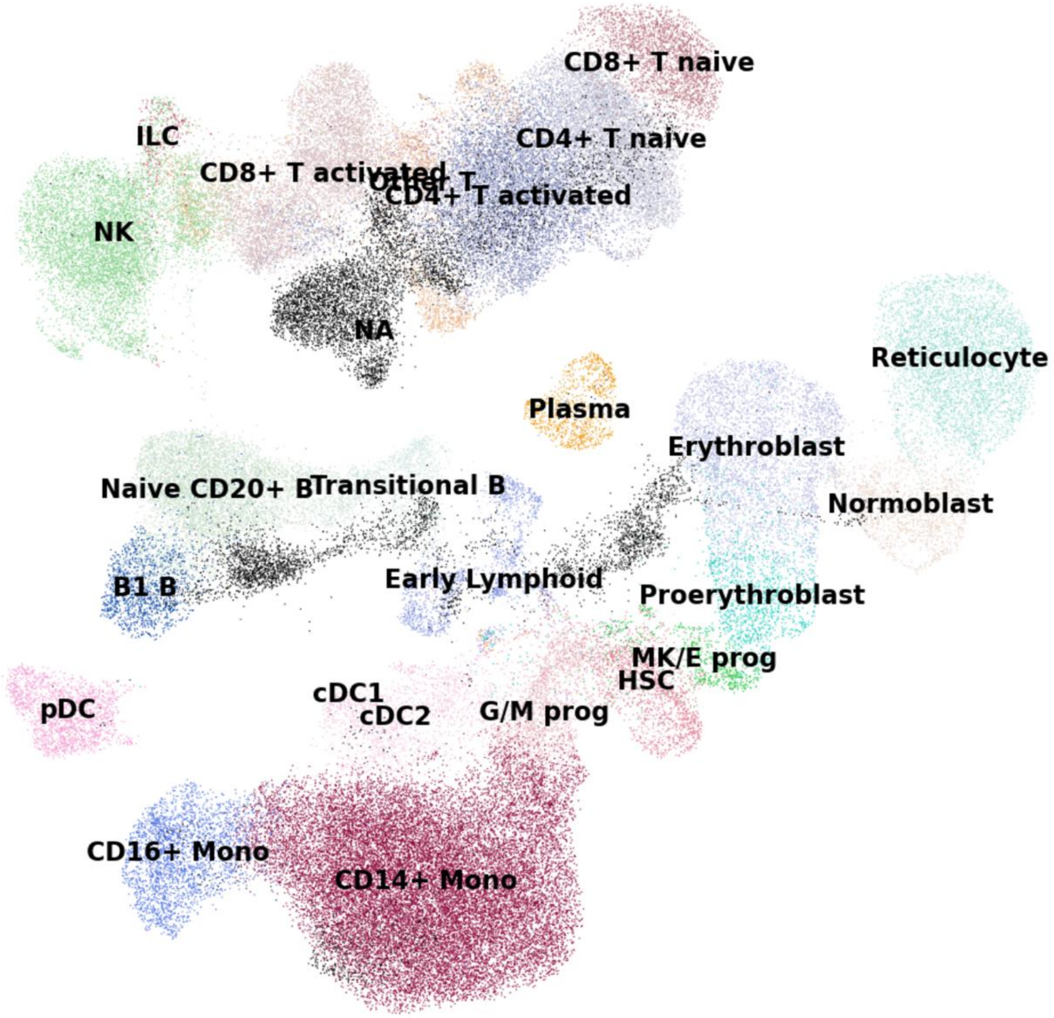


Heumos, L., Schaar, A.C., Lance, C. et al. Best practices for single-cell analysis across modalities. Nat Rev Genet 24, 550–572 (2023). https://doi.org/10.1038/s41576-023-00586-w

Recap: Where we stand after a whole lot of processing

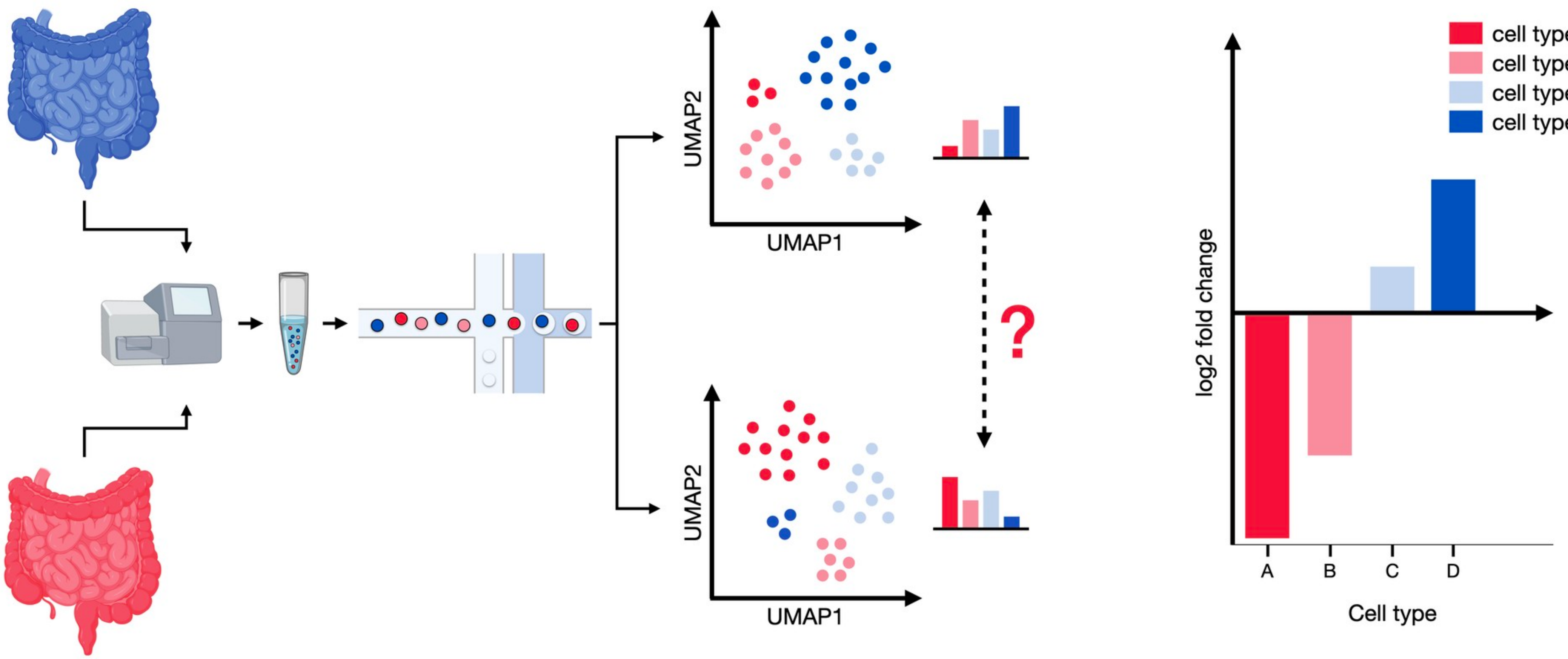


Finally, an annotated dataset.



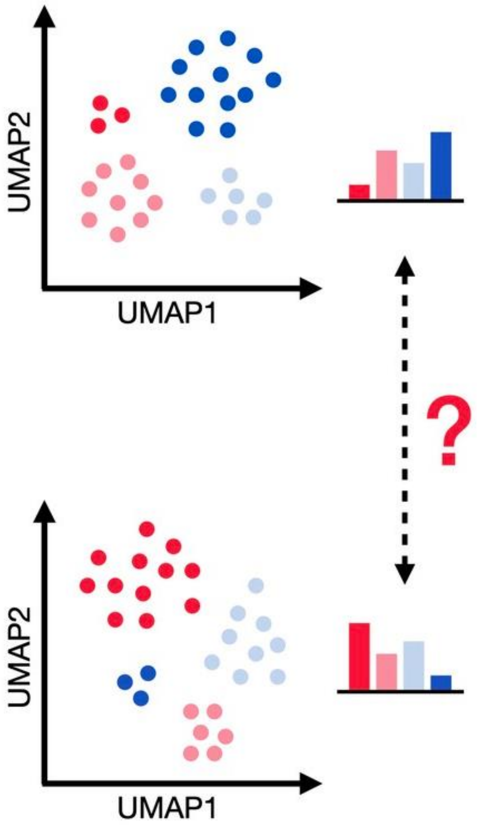
https://www.sc-best-practices.org/cellular_structure/annotation.html

What to do with an annotated dataset:
1) Compositional analysis



https://www.sc-best-practices.org/conditions/compositional.html

Cell type fraction data is compositional.

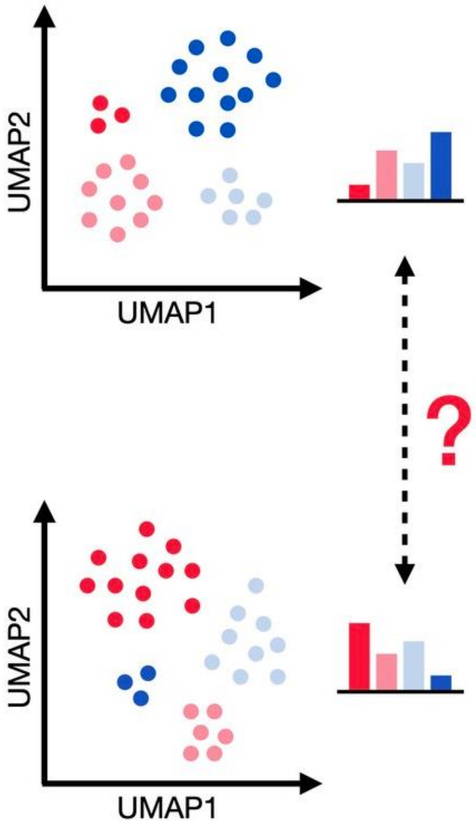


Type A	5%
Type B	30%
Type C	25%
Type D	40%

Sums to 100%

https://www.sc-best-practices.org/conditions/compositional.html

Cell type fraction data is compositional.



Type A	5%
Type B	30%
Type C	25%
Type D	40%

Sums to 100%

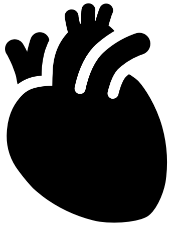
If one cell type becomes more abundant,

the fractional contribution of the other cell types goes down

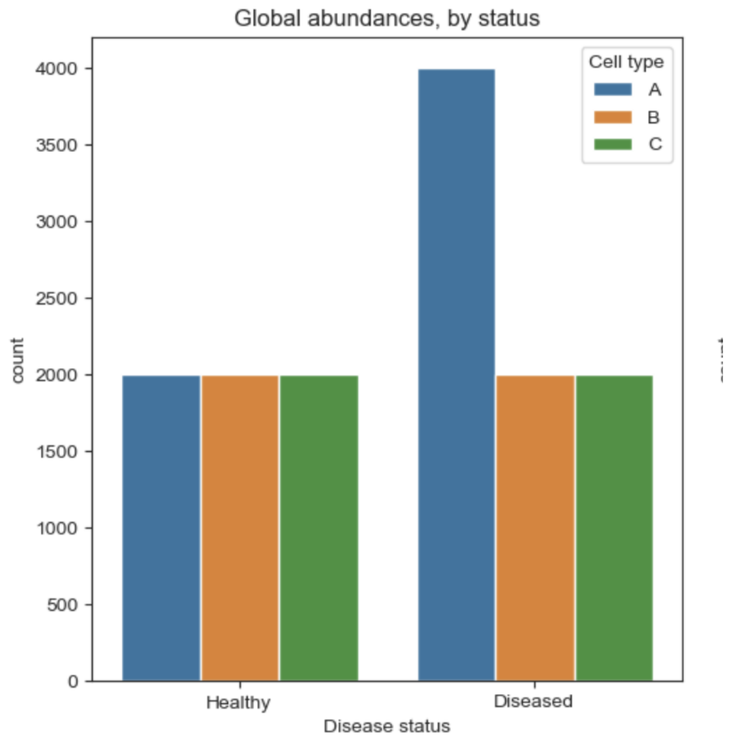
even though their absolute numbers may be unchanged.

https://www.sc-best-practices.org/conditions/compositional.html

Cell type count data **from a sample** is compositional.



Whole organ

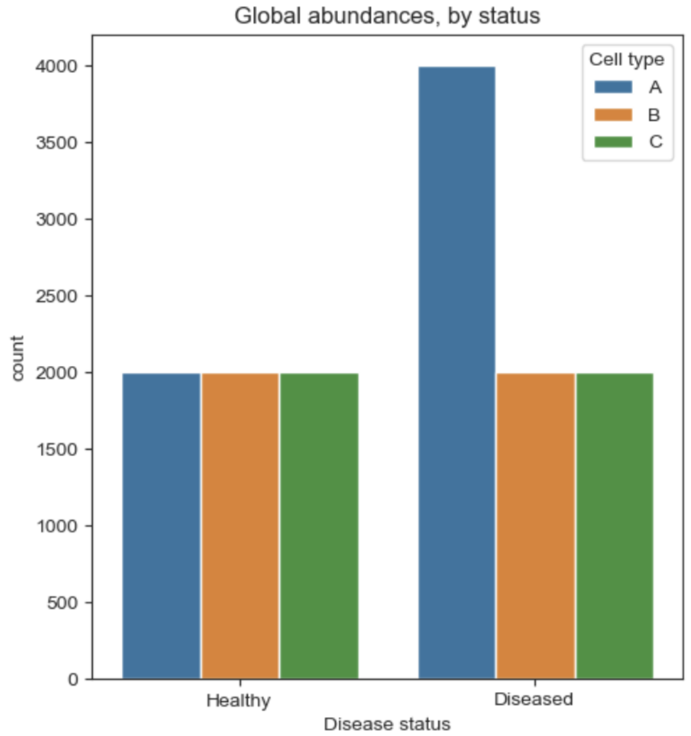


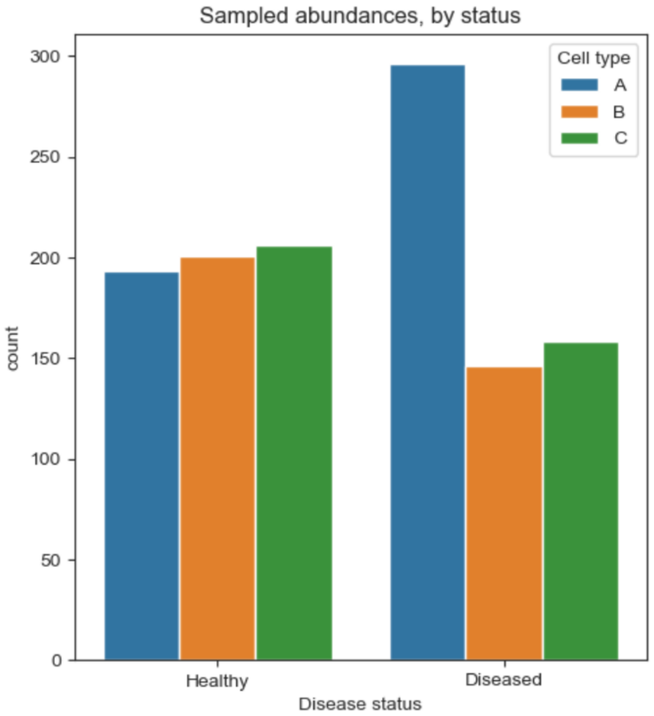
https://www.sc-best-practices.org/conditions/compositional.html

Cell type count data **from a sample** is compositional.

Whole organ

small sample
from organ





https://www.sc-best-practices.org/conditions/compositional.html

Compositional data requires special statistical methods.

Compositional data is characterized by inherent negative correlations between its features

[If one goes up, another must go down]

Compositional data requires special statistical methods.

Compositional data is characterized by inherent negative correlations between its features

[If one goes up, another must go down]

Statistical methods for non-compositional data (e.g. t-test, Wilcoxon's test…) may return false positive results when testing for differential abundance

Compositional data requires special statistical methods.

Compositional data is characterized by inherent negative correlations between its features

[If one goes up, another must go down]

Statistical methods for non-compositional data (e.g. t-test, Wilcoxon's test...) may return false positive results when testing for differential abundance

ANNUAL REVIEW OF STATISTICS AND ITS APPLICATION Volume 8, 2021

Review Article | Free

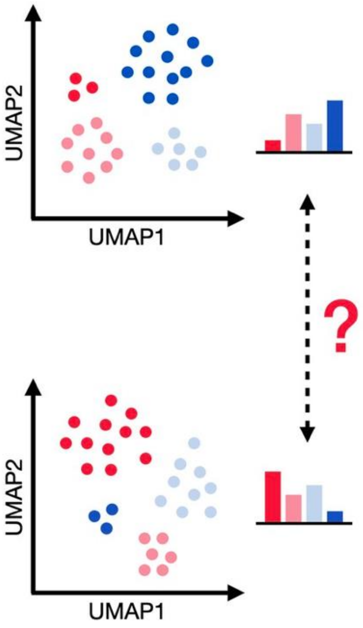
Compositional Data Analysis

Michael Greenacre[1]

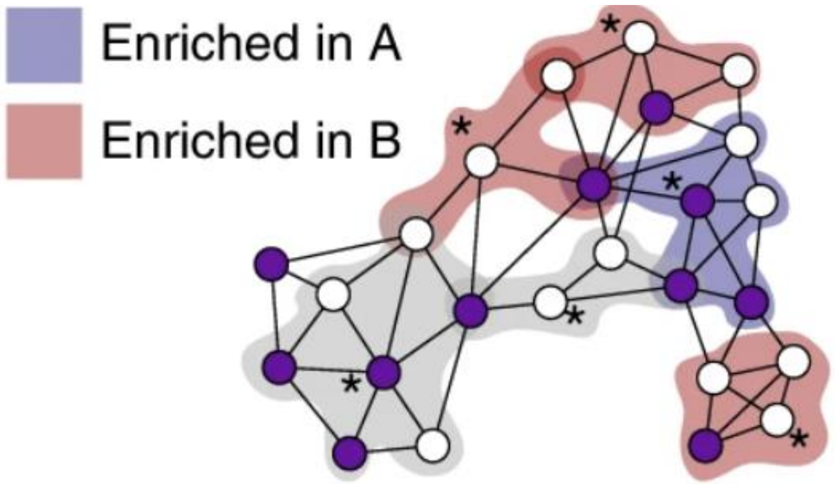
https://www.annualreviews.org/content/journals/10.1146/annurev-statistics-042720-124436

For differential abundance testing in sc data, use dedicated methods.

scCODA for labelled clusters



miloR for graph neighborhoods (does not need labels)



https://www.nature.com/articles/s41467-021-27150-6

https://www.nature.com/articles/s41587-021-01033-z

For differential abundance testing in sc data, use dedicated methods.

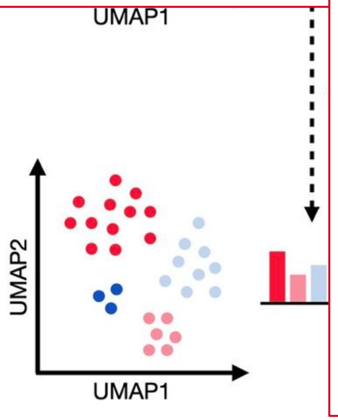
scCODA for labelled clusters

miloR for graph neighborhoods
ot need labels)

RESEARCH ARTICLE | BIOPHYSICS AND COMPUTATIONAL BIOLOGY

sccomp: Robust differential composition and variability analysis for single-cell data

Stefano Mangiola, Alexandra J. Roth-Sch
Info & Affiliations



Method | Open access | Published: 26 June 2023

DCATS: differential composition analysis for flexible single-cell experimental designs

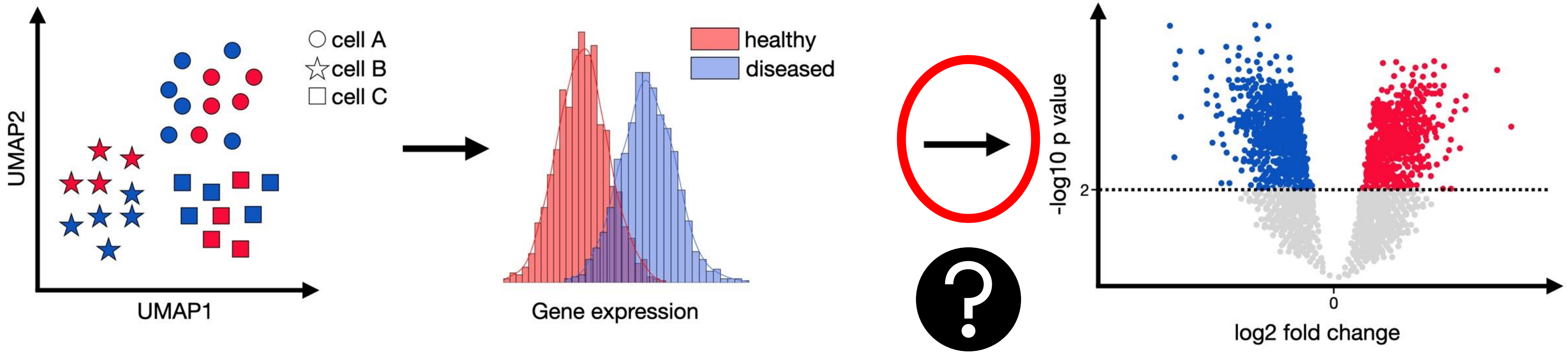
Xinyi Lin, Chuen Chau, Kun Ma, Yuanhua Huang & Joshua W. K. Ho

Genome Biology **24**, Article number: 151 (2023) | Cite this article

https://www.nature.com/articles/s41467-021-27150-6

https://www.nature.com/articles/s41587-021-01033-z

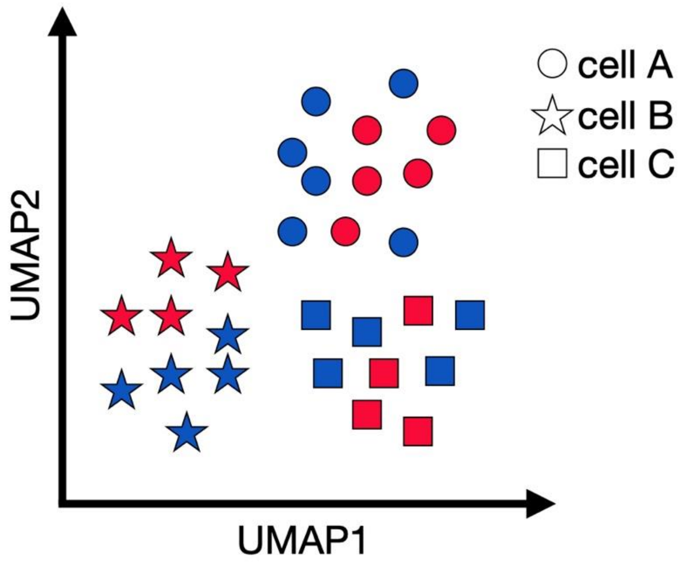
What to do with an annotated dataset:
2) Differential expression testing between conditions



https://www.sc-best-practices.org/conditions/differential_gene_expression.html

Differential Gene Expression Testing: Cells are not independent replicates

Aim: compare case and control, e.g. disease X vs. healthy



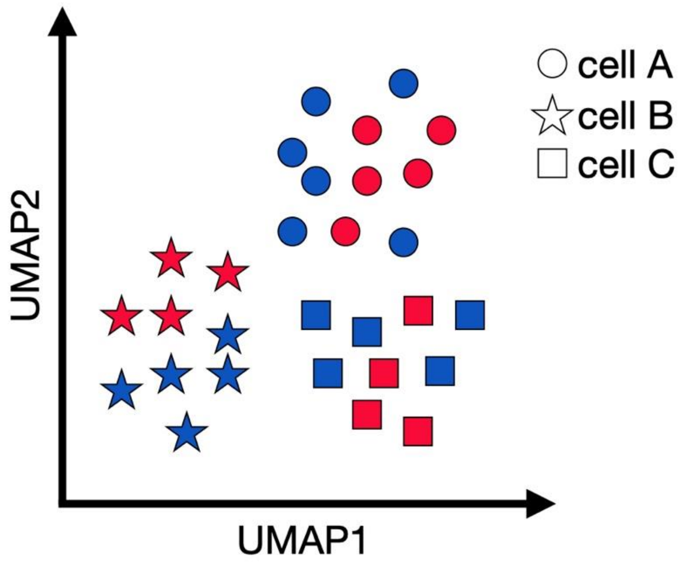
Many highly significant results!

Naïve approach
Statistical comparison (e.g. Wilcoxon via marker gene tests) between celltype A in case and control

https://www.sc-best-practices.org/conditions/differential_gene_expression.html

Differential Gene Expression Testing: Cells are not independent replicates

Aim: compare case and control, e.g. disease X vs. healthy



Naïve approach
Statistical comparison (e.g. Wilcoxon via marker gene tests) between celltype A in case and control

Many highly significant results!

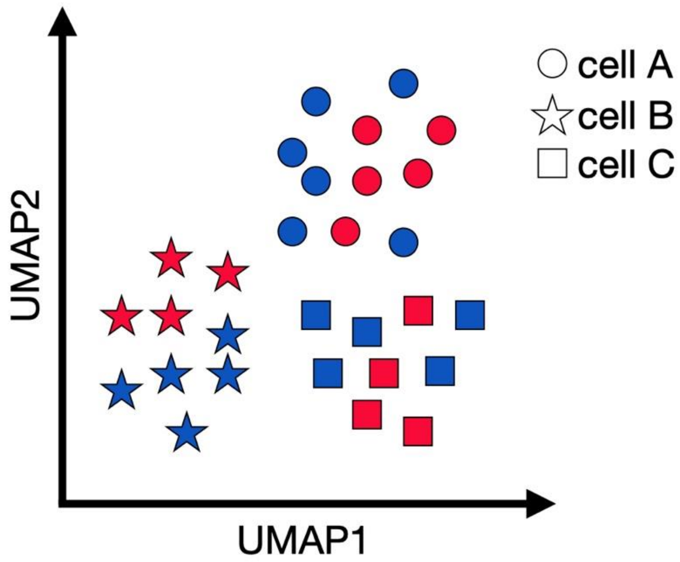
But: Treats each cell as an independent replicate, ignoring that cells from same patient are correlated!

https://www.sc-best-practices.org/conditions/differential_gene_expression.html

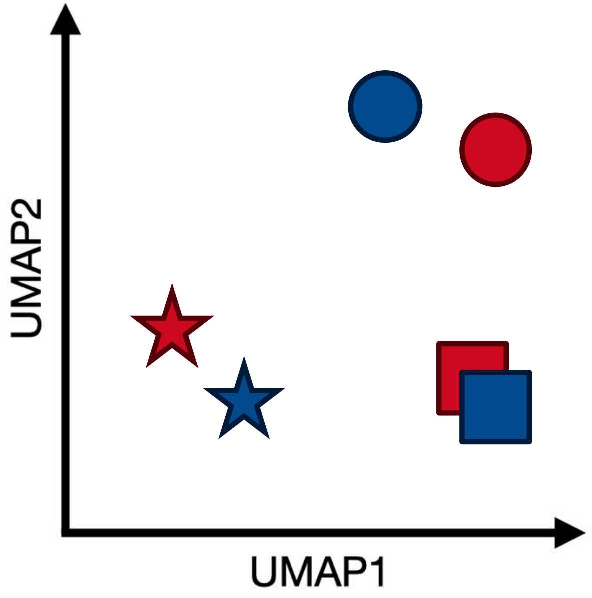
Differential Expression – Marker Testing vs Condition comparison

Marker gene testing (within dataset)	**Differential expression between conditions**
Compares: "Cluster X" vs "all other cells"	Compares: "Condition A" vs "Condition B" within same cell type
Purpose: Identify cell type-defining features	Purpose: Identify condition-responsive genes
Tests assume cells are independent observations (i.e. each cell is considered a replicate)	- Requires biological replicates (multiple patients/samples) - Provides ability to account for donor-specific effects (paired samples) and other covariates
FindMarkers() and friends	**DEG framework like DESeq2, edgeR and friends**

One valid strategy for to avoid each cell being considered an independent replicate: Pseudobulking



mean
or
sum
of
counts



- One gene expression vector per sample and cell type
- Less sparse

https://www.sc-best-practices.org/conditions/differential_gene_expression.html

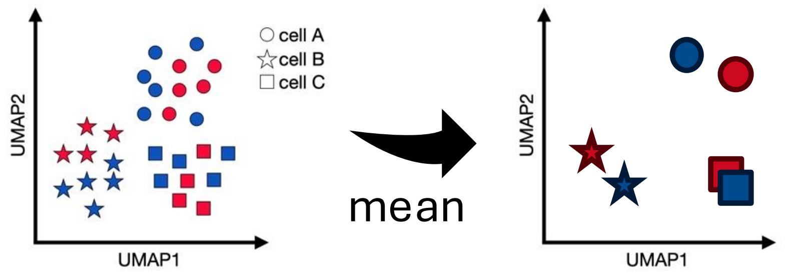
Overview: Approaches to sc DEG testing

pseudobulk methods	single-cell specific methods

Overview: Approaches to sc DEG testing

pseudobulk methods

1. Preprocess, cluster and annotate the dataset
2. Aggregate counts by taking the mean per cell type and sample/patient
3. Apply a differential expression method like for bulk data



single-cell specific methods

https://www.sc-best-practices.org/conditions/differential_gene_expression.html

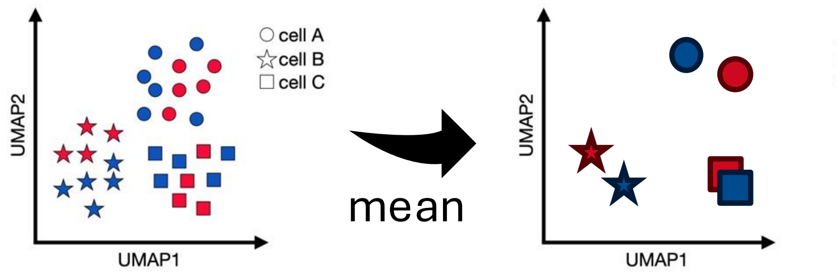
Overview: Approaches to sc DEG testing

pseudobulk methods

1. Preprocess, cluster and annotate the dataset
2. Aggregate counts by taking the mean per cell type and sample/patient
3. Apply a differential expression method like for bulk data

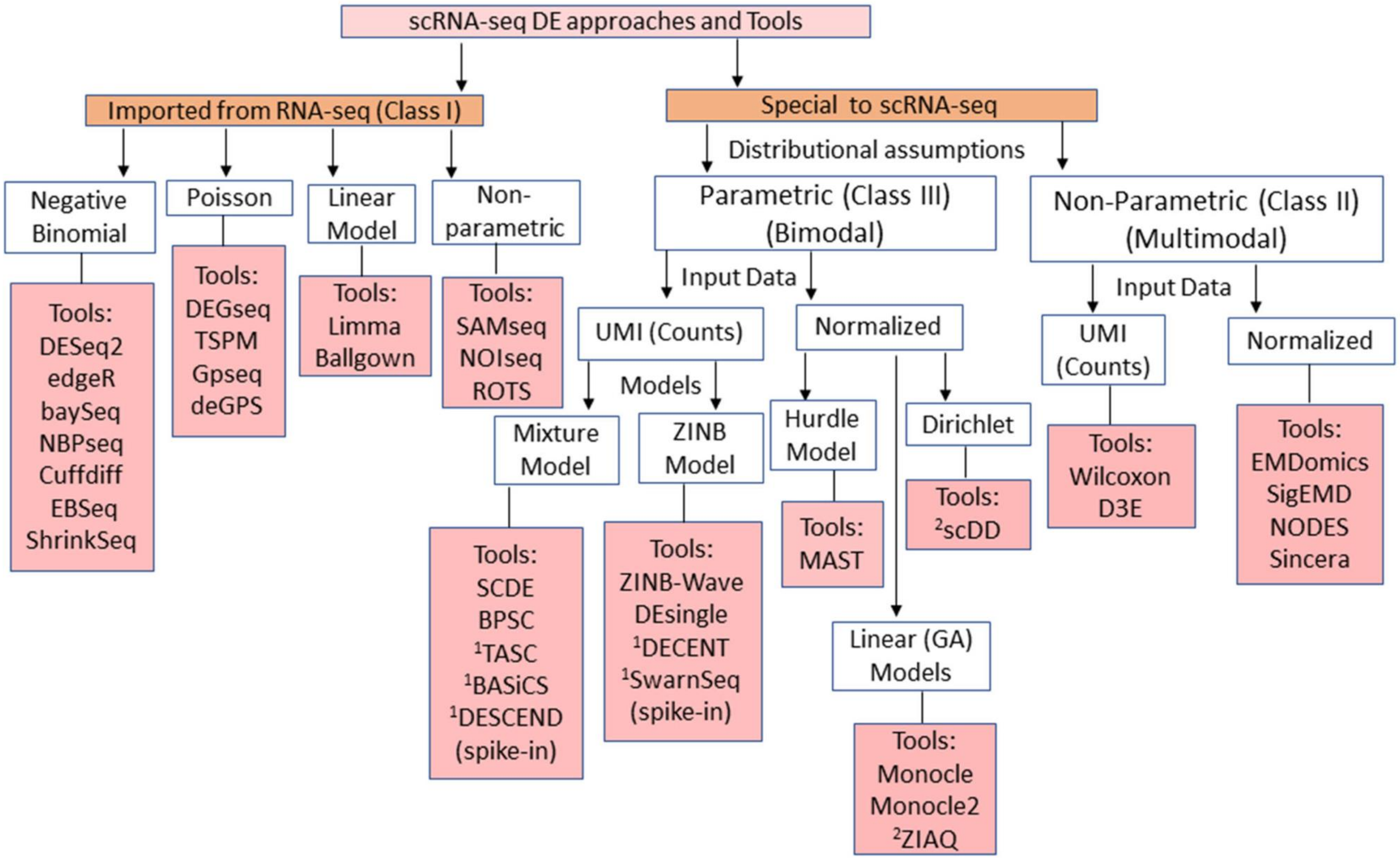
single-cell specific methods

- Typically use generalized mixed effects models
- Model specific single-cell data noise properties accurately



https://www.sc-best-practices.org/conditions/differential_gene_expression.html

Overview: Approaches to sc DEG testing



Das, S. et al. A Comprehensive Survey of Statistical Approaches for Differential Expression Analysis in Single-Cell RNA Sequencing Studies. Genes 2021, 12, 1947. https://doi.org/10.3390/genes12121947

Overview: Approaches to sc DEG testing

pseudobulk methods

single-cell specific methods

!

Consensus and robustness across methods are low!

Squair, J.W., Gautier, M., Kathe, C. et al. Confronting false discoveries in single-cell differential expression. Nat Commun 12, 5692 (2021). https://doi.org/10.1038/s41467-021-25960-2

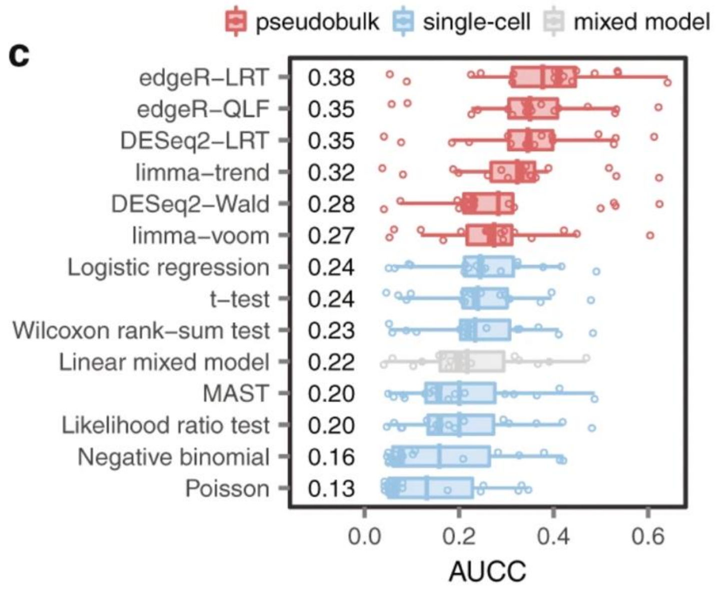
Overview: Approaches to sc DEG testing

pseudobulk methods

single-cell specific methods

!

Consensus and robustness across methods are low!



Pseudobulk methods perform favourably against single-cell specific methods.

Squair, J.W., Gautier, M., Kathe, C. et al. Confronting false discoveries in single-cell differential expression. Nat Commun 12, 5692 (2021). https://doi.org/10.1038/s41467-021-25960-2

Example: pseudobulk/DESeq2
1) Create pseudobulk object

Example dataset: *in vitro* stimulated PBMCs from 8 Lupus patients before and after 6h-treatment with IFN-β (16 samples in total)

label	cluster	cell_type	replicate
ctrl	9	CD14+ Monocytes	patient_1016
ctrl	9	CD14+ Monocytes	patient_1256
ctrl	3	CD4 T cells	patient_1488
ctrl	9	CD14+ Monocytes	patient_1256
ctrl	4	Dendritic cells	patient_1039

annotated single cell dataset with **raw counts** [thousands of rows]

calculate summed gene expression per patient and cell type

→

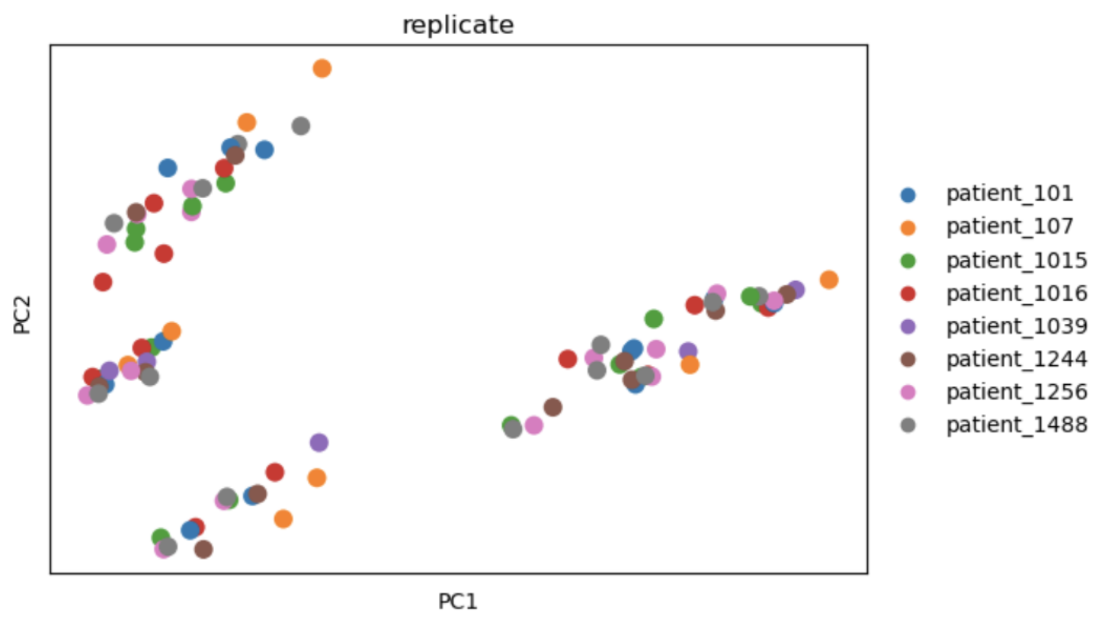
Pseudobulk dataset with one entry per patient_celltype [patient x cell types rows] [e.g. 16 x 7 = 112]

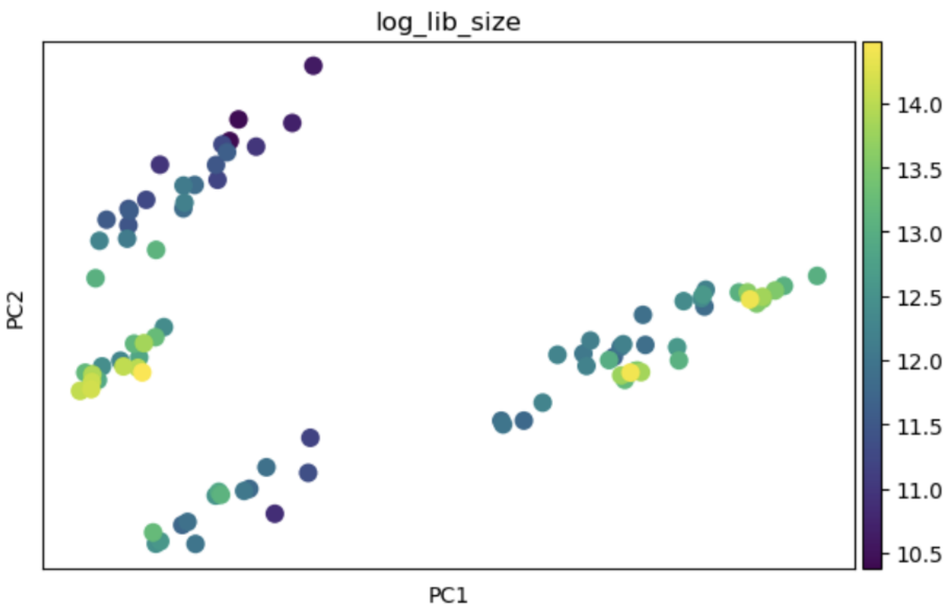
https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk

Example: pseudobulk/ DESeq2
2) Inspect major axes of variation

The statistical model used for differential gene expression must capture **major axes of variation** to return accurate differential gene expression results.

Lognormalize pseudobulk data → PCA → inspect covariates



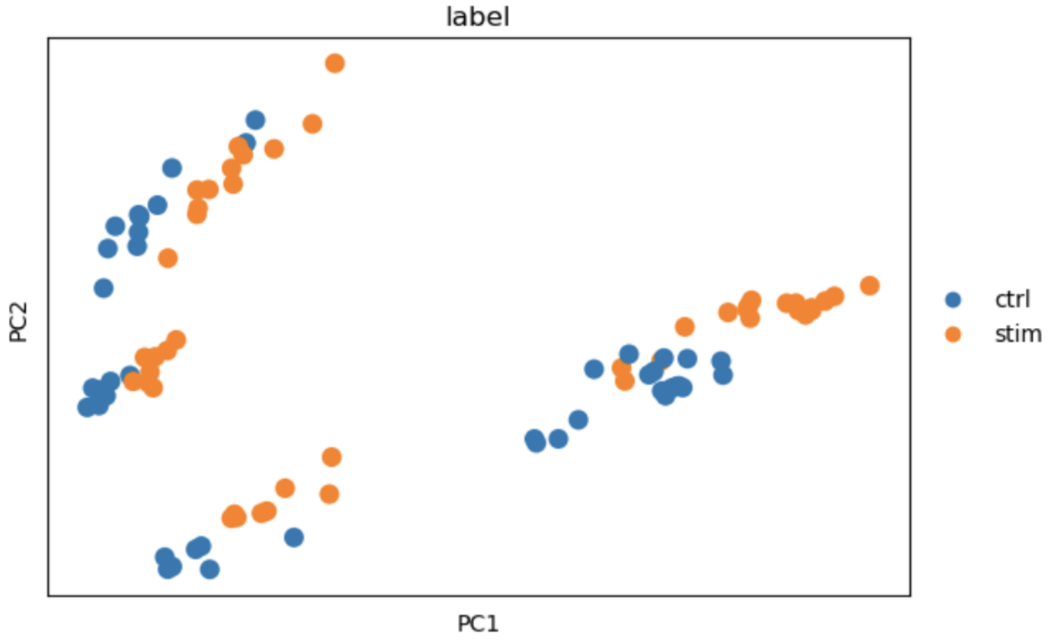


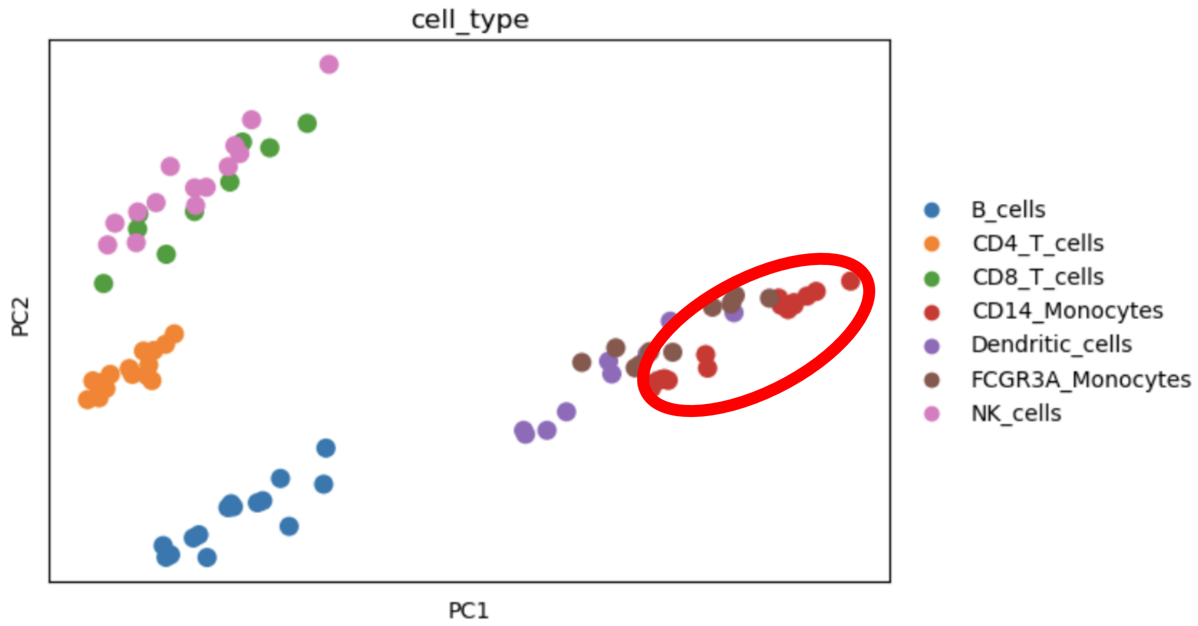
https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk

Example: pseudobulk/ DESeq2
2) Inspect major axes of variation

The statistical model used for differential gene expression must capture **major axes of variation** to return accurate differential gene expression results.

Lognormalize pseudobulk data → PCA → inspect covariates





https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk

Example: pseudobulk/ DESeq2
3) Run a differential expression test for a chosen cell type

Since we identified no major confounders during the exploratory analysis, we set up the simplest **design matrix** with the stimulation label as sole covariate.

design matrix

Sample	Intercept	condition	Interpretation
ctrl_1	1	0	baseline expression
ctrl_2	1	0	baseline expression
ctrl_3	1	0	baseline expression
stim_1	1	1	baseline + treatment
stim_2	1	1	baseline + treatment
stim_3	1	1	baseline + treatment

Example: pseudobulk/ DESeq2
3) Run a differential expression test for a chosen cell type

Since we identified no major confounders during the exploratory analysis, we set up the simplest **design matrix** with the stimulation label as sole covariate.

design matrix

Sample	Intercept	condition	Interpretation
ctrl_1	1	0	baseline expression
ctrl_2	1	0	baseline expression
ctrl_3	1	0	baseline expression
stim_1	1	1	baseline + treatment
stim_2	1	1	baseline + treatment
stim_3	1	1	baseline + treatment

model

$$\log(\text{expected counts}) = \beta_0 \times \text{Intercept} + \beta_1 \times \text{condition}$$

Example: pseudobulk/ DESeq2
3) Run a differential expression test for a chosen cell type

Since we identified no major confounders during the exploratory analysis, we set up the simplest **design matrix** with the stimulation label as sole covariate.

design matrix

Sample	Intercept	condition	Interpretation
ctrl_1	1	0	baseline expression
ctrl_2	1	0	baseline expression
ctrl_3	1	0	baseline expression
stim_1	1	1	baseline + treatment
stim_2	1	1	baseline + treatment
stim_3	1	1	baseline + treatment

model

$$\log(\text{expected counts}) = \beta_0 \times \text{Intercept} + \beta_1 \times \text{condition}$$

Is β_1 (the log2 fold change) different from 0?

Example: pseudobulk/ DESeq2
3) Run a differential expression test for a chosen cell type

Since we identified no major confounders during the exploratory analysis, we set up the simplest **design matrix** with the stimulation label as sole covariate.

design matrix

Sample	Intercept	condition	Interpretation
ctrl_1	1	0	baseline expression
ctrl_2	1	0	baseline expression
ctrl_3	1	0	baseline expression
stim_1	1	1	baseline + treatment
stim_2	1	1	baseline + treatment
stim_3	1	1	baseline + treatment

model

$$\log(\text{expected counts}) = \beta_0 \times \text{Intercept} + \beta_1 \times \text{condition}$$

Is β_1 (the log2 fold change) different from 0?

What packages like DESeq2 do under the hood:

Size factor estimation: Normalizes for library size differences between samples

Dispersion estimation: Models the relationship between mean expression and variance across genes

Statistical testing: Fits negative binomial generalized linear models and performs Wald tests

Pseudobulk DEG analysis in practice

Seurat (V5+) comes with inbuilt pseudobulking capabilities, for DEG calculation we use **DESeq2**

```{r run_DEseq2}
Idents(kang2018_pb_monocytes) <- "label"

bulk_monocytes_de <- FindMarkers(object = kang2018_pb_monocytes,
                                 ident.1 = "ctrl",
                                 ident.2 = "stim",
                                 test.use = "DESeq2")
```

In **scanpy**, we make use of the additional libraries **decoupler** and **PyDESeq2** (slightly more complicated, but also more flexible)

```
inference = DefaultInference(n_cpus=None)
dds = DeseqDataSet(
    counts=counts_df,
    metadata=metadata_df,
    design="~label",
    refit_cooks=True,
    inference=inference,
)
```

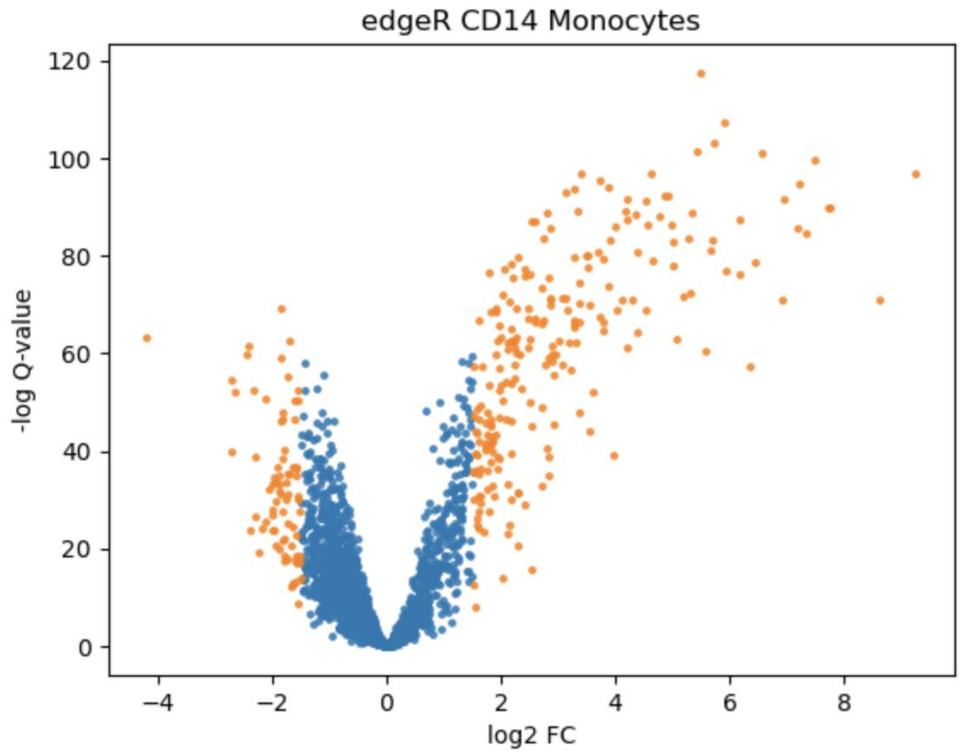

https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk

Inspect and visualize DEG results: Volcano plot

CD14+
monocytes

	logFC	logCPM	F	PValue	FDR
HESX1	8.345536	6.773420	1281.013295	1.837373e-15	2.766927e-12
CD38	7.126846	7.420668	1243.793133	2.266164e-15	2.766927e-12
NT5C3A	5.657050	8.327003	1218.102628	2.628780e-15	2.766927e-12
SOCS1	4.388247	6.943768	1191.289806	3.079524e-15	2.766927e-12
GMPR	6.943484	7.031832	1159.601183	3.730018e-15	2.766927e-12

Filter for genes with FDR < 0.01 And logFC>1.5, (here marked in orange)



https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk

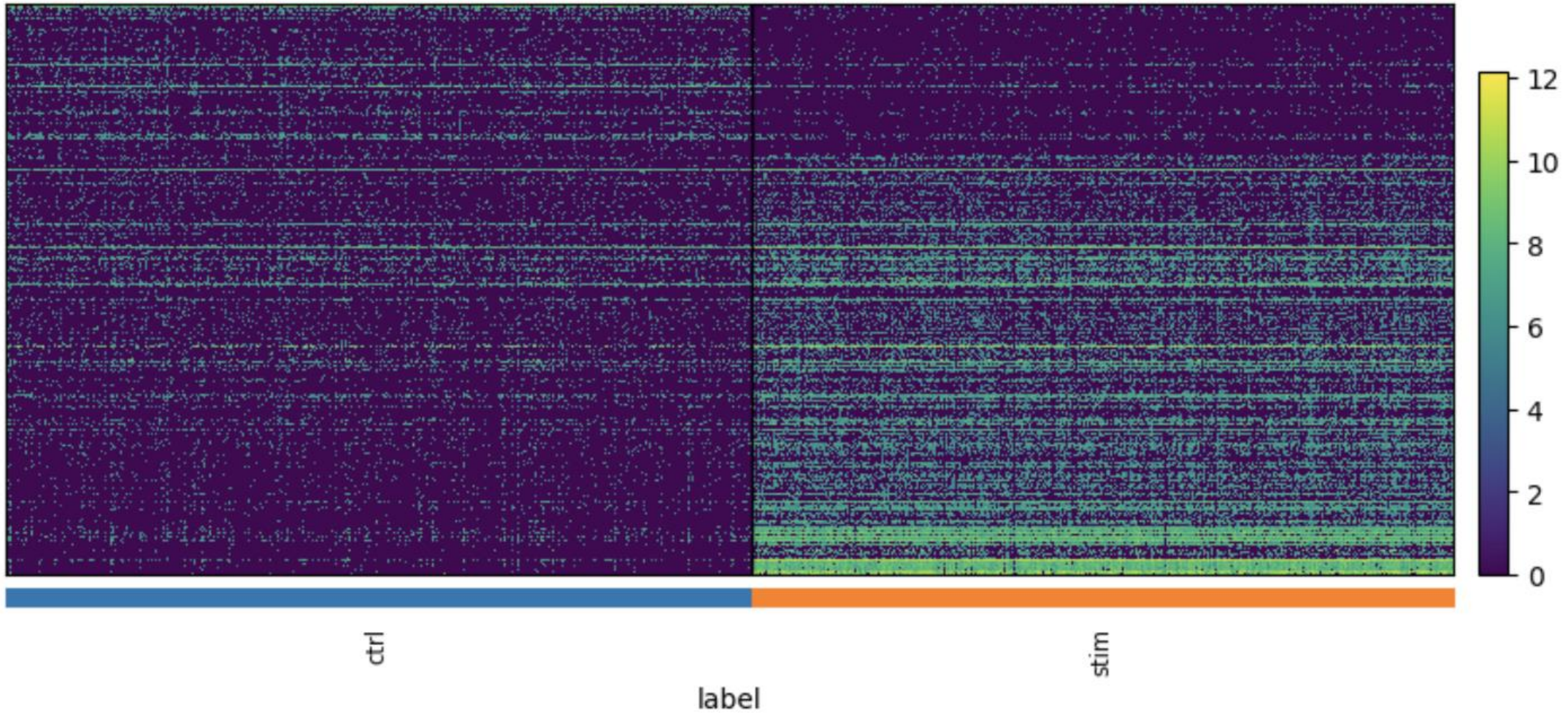
Inspect and visualize DEG results: Heatmap

CD14+ monocytes

Filter for genes with FDR < 0.01 And logFC>1.5

Cells →

Genes →



https://www.sc-best-practices.org/conditions/differential_gene_expression.html#pseudobulk